Discrete Mathematics Homework #2

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Question no21

$35-(\text{chapter no} \times \text{last 3 digits of my num}) \bmod 35$

$35 - (7 \times 017) \bmod 35 = 21$

Question

Consider the following game. A person flips a coin repeatedly until a head comes up. This person receives a payment of 2^n dollars if the first head comes up at the nth flip.

a) Let X be a random variable equal to the amount of money the person wins. Show that the expected value of X doesn't exist (that is, it is infinite). Show that a rational gambler, that is, someone willing to pay to play the game as long as the price to play is not more than the expected payoff, should be willing to wager any amount of money to play this game.

solution (a)

$P(X = 2^n) = \frac{1}{2^n}$ and $P(X = k \neq 2^n) = 0$, therefore

$$E[X] = \sum_{n=1}^{\infty} 2^n \cdot \frac{1}{2^n} = \infty$$

if a gambler wants to wager assuming he wins $\Rightarrow$ after N flips, then he should be ready to pay up to N dollars which is not a bounded function.

b) Suppose that the person receives 2^n dollars if the first head comes up on the nth flip where $n < 8$ and $2^8 = 256$ dollars if the first head comes up on or after the eighth flip. What is the expected value of the amount of money the person wins? How much money should a person be willing to pay to play this game

solution (b)

$$E[X] = \sum_{n=1}^{8} 2^n \cdot \frac{1}{2^n} + \sum_{n=9}^{\infty} 256 \cdot \frac{1}{2^n} = 9 \text{ dollars.}$$

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Question no 4

$35 - (\text{chapter no} \times \text{last 2 digit of my num}) \bmod 35$

$35 - (8 \times 017) \bmod 35 = 4$

Question

Suppose that every hour there are two new bacteria in a colony for each bacterium that was present the previous hour, and that all bacteria 2 hour old die. The colony starts with 100 new bacteria.

a) Set up a recurrence relation for the number of bacteria present after n hours.

solution (a)

Let a_n be the number of bacteria present after n hour. At the nth hour a_{n-1} bacteria present hour and a_{n-2} bacteria are 2 hours old. $a_{n-1} - a_{n-2}$ bacteria will still be present at the nth hour.

Each of the a_{n-1} bacteria present in the previous hour product 2 new bacteria. $\Rightarrow 2a_{n-1}$

a_n is the sum of the number of bacteria in each step! $a_n = a_{n-1} - a_{n-2} + 2a_{n-1} = 3a_{n-1} - a_{n-2}$

$a_0 = 100 \qquad a_1 = 100 + 200 = 300$

b) What is the solution of this recurrence solution?

solution (b)

Let $a_n = r^2$, $a_{n-1} = r$ and $a_{n-2} = 1$

$r^2 = 3r - 1 \rightarrow r^2 - 3r + 1 = 0 \rightarrow r = \frac{3 \pm \sqrt{3^2 - 4(1)(1)}}{2}$, $r = \frac{3 \pm \sqrt{5}}{2}$

the solution of the recurrence relation: $a_n = a_1 r_1^n + a_2 r_2^n$

$a_n = a_1 \left(\frac{3-\sqrt{5}}{2}\right)^n + \left(\frac{3-\sqrt{5}}{2}\right)^n \cdot a_2$, $100 = a_1 + a_2$, $300 = \left(\frac{3-\sqrt{5}}{2}\right) a_1 + \left(\frac{3+\sqrt{5}}{2}\right) a_2$

$50(3-\sqrt{5}) = \left(\frac{3-\sqrt{5}}{2}\right) a_1 + \left(\frac{3+\sqrt{5}}{2}\right) a_2$

$\Leftrightarrow a_2 = 50 + 30\sqrt{5}$

$a_1 = 50 - 30\sqrt{5}$

$a_n = (50 - 30\sqrt{5}) \cdot \left(\frac{3-\sqrt{5}}{2}\right)^n + (50 + 30\sqrt{5}) \cdot \left(\frac{3+\sqrt{5}}{2}\right)^n$

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Question no 22

$35 - (\text{chapter no}) \times (\text{last 3 digits of my num}) \bmod 35$

$35 - (9 \times 017) \bmod 35 = 22$

Question

Show that $\{(x,y) \mid x-y \in \mathbb{Q}\}$ is an equivalence relation on the set of real numbers, where $\mathbb{Q}$ denotes the set of rational numbers. What are $[1]$, $[\frac{1}{2}]$, and $[\pi]$?

Solution

$R = \{(x,y) \mid x-y \in \mathbb{Q}\}$

$A = \mathbb{R}$

Reflexive Let $x \in \mathbb{R}$! $x - x = 0 \in \mathbb{Q}$

Thus $(x,x) \in R$ and Thus R is reflexive

Symmetric Let $(a,b) \in R$ $a-b \in \mathbb{Q}$ also $-a+b \in \mathbb{Q}$, $= b-a \in \mathbb{Q}$

which implies $(b,a) \in R$ and thus R is symmetric.

Transitive Let $(a,b) \in R$ and $(b,c) \in R$

$a-b \in \mathbb{Q}$

$b-c \in \mathbb{Q}$, The sum of two rational numbers is also a rational number

$(a-b)+(b-c) \in \mathbb{Q}$ so $a-c \in \mathbb{Q}$.

which implies $(a,c) \in R$ and thus R is transitive.

Conclusion R is an equivalence relation, because R is Reflexive symmetric and transitive.

$[1] = \{y \mid 1-y \in \mathbb{Q}\} = \{y \mid y \in \mathbb{Q}\} = \mathbb{Q}$

$[\frac{1}{2}] = \{y \mid \frac{1}{2} - y \in \mathbb{Q}\} = \{y \mid y \in \mathbb{Q}\} = \mathbb{Q}$

$[\pi] = [y \mid \pi - y \in \mathbb{Q}\} = \{\pi + r \mid r \in \mathbb{Q}\}$

Discrete mathematics Homework #2

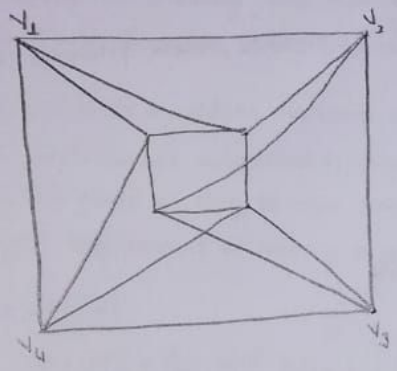
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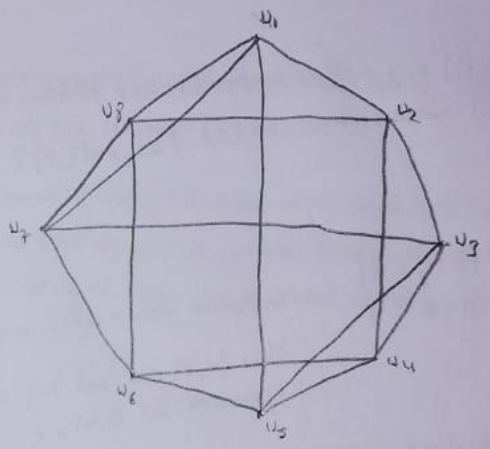
Question no 5

$35 - (\text{chapter no} \times \text{last 3 digits of my num}) \bmod 35$

$35 - (10 \times 017) \bmod 35 = 5$

Question





The comlete m-partite graph $K_{n_1, n_2, \ldots n_m}$ has vertices partitioned into m subsets of $n_1, n_2, \ldots n_m$ elements each, and vertices are adjacent if they are in different subsets in the partition.

Solution

These graphs are isomorphic. Looking at the size of the regions we can find an isomorphism. One such correspondence is $u_1 \Leftrightarrow v_5$, $u_2 \Leftrightarrow v_4$, $u_3 \Leftrightarrow v_7$, $u_4 \Leftrightarrow v_3$, $u_5 \Leftrightarrow v_8$, $u_6 \Leftrightarrow v_2$, $u_7 \Leftrightarrow v_6$ and $u_8 \Leftrightarrow v_1$. We just need to check that all the edges are preserved by this correspondence.

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Question no 23

$35 - (\text{chapter no} \times \text{last 3 digits of my num}) \bmod 35$

$35 - (11 \times 017) \bmod 35 = 23$

Question

Devise an algorithm for listing the vertices of an ordered rooted tree in level order.

Solution

```
procedure level order (T=ordered rooted tree with root r)
queue = r
list  = [] -> empty list
while queue is not empty
    v = first vertex
    list = list, v
    queue = queue with v removed and all children of v added at the end of queue
    return list
```